

Introduction to Mathematics and Optimisation

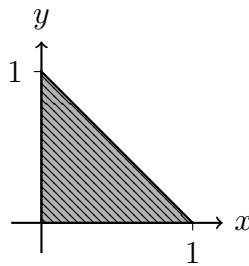
Exercises Week 1

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1 Exercise 1.58

Consider the subset S of $\mathbb{R}^2$ pictured in the drawing below.



Express S as $S = \{(x, y) \in \mathbb{R}^2 \mid p_1(x, y), p_2(x, y), p_3(x, y)\}$ where p_1, p_2, p_3 are predicates in the variables x, y .

Express $\mathbb{R}^2 \setminus S$ as $\{(x, y) \in \mathbb{R}^2 \mid p(x, y)\}$ where $p(x, y) = q_1(x, y) \vee q_2(x, y) \vee q_3(x, y)$ and q_1, q_2 and q_3 are suitable predicates in the variables x, y .

1.1 Solution

We can solve this problem by defining what we know about the drawing

- All x values are between 0 and 1 included
- All y values are between 0 and 1 included
- All points have y coordinate below the line formed by points $[0, 1]$ and $[1, 0]$ including the line

Each of those items can be written as predicate the first two are easy:

- $p_1(x, y) : 0 \leq x \leq 1$
- $p_2(x, y) : 0 \leq y \leq 1$

The third one can be formalized by taking the equation forming this line and rewriting it as an inequality

$$\begin{aligned}y &= -x + 1 \\y &\leq -x + 1 \\x + y &\leq 1\end{aligned}$$

So the third predicate is

- $p_3(x, y) : x + y \leq 1$

This makes the solution for the first part

$$S = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1 \wedge 0 \leq y \leq 1 \wedge x + y \leq 1\}$$

The second part is asking us to define the $\mathbb{R}^2 \setminus S$ which is defined by inversion of the the three predicates forming S which can be written as this $\neg(0 \leq x \leq 1 \wedge 0 \leq y \leq 1 \wedge x + y \leq 1)$. Using De Morgan's law we can rewrite it as $\neg(0 \leq x \leq 1) \vee \neg(0 \leq y \leq 1) \vee \neg(x + y \leq 1)$ the inverting the conditions gives us the final solution

$$\mathbb{R}^2 \setminus S = \{(x, y) \in \mathbb{R}^2 \mid (x < 0 \vee x > 1) \vee (y < 0 \vee y > 1) \vee (x + y > 1)\}$$

2 Exercise 1.79

Assume that

$$\frac{a}{b} < \frac{c}{d}$$

Write down a precise argument showing that

$$\frac{a+c}{b+d} < \frac{c}{d}$$

2.1 Proof

from the statement $\frac{a}{b} < \frac{c}{d}$ we can deduce some useful things

$$\begin{aligned}\frac{a}{b} < \frac{c}{d} &\Rightarrow \\b > 0 \wedge d > 0 &\Rightarrow \\b + d > 0 &\end{aligned}$$

this tells us it's safe to multiply the inequality by b, d and $b + d$ we can also find equivalent statement to the original one

$$\frac{a}{b} < \frac{c}{d} \Leftrightarrow \\ ad < bc$$

$$\begin{aligned} \frac{a+c}{b+d} &< \frac{c}{d} \Leftrightarrow \\ d(a+c) &< c(b+d) \text{ (cross multiplication)} \Leftrightarrow \\ da+dc &< cb+cd \text{ (distributive law)} \Leftrightarrow \\ da &< cb \Leftrightarrow \\ ad &< bc \text{ (commutative law)} \end{aligned}$$

We have ended with the statement $ad < bc$ which is equivalent to $\frac{a}{b} < \frac{c}{d}$ and since we know this is equivalent we can be sure that assuming $\frac{a}{b} < \frac{c}{d}$ then $\frac{a+c}{b+d} < \frac{c}{d}$ is true

2.2 LEAN

The proof can also be formalized using the programming language LEAN. I'm not including it as my main proof since I constructed this one with the help of artificial intelligence.

```
import Mathlib
```

```
theorem mediant_inequality_with_rationals
```

```
  (a b c d : ℤ) (hb : (0 : ℚ) < b) (hd : (0 : ℚ) < d) (h : (a : ℚ) / b <
    ((a + c) : ℚ) / (b + d) < (c : ℚ) / d := by
  have hbd : (0 : ℚ) < b + d := add_pos hb hd
```

```

  rw [div_lt_div_iff₀ hbd hd]
  rw [div_lt_div_iff₀ hb hd] at h
  linarith
```